SAMPLE

# Earth's Sky: Earthshine and Earthlight

The Glow on the Moon's Dark Side

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## Introduction

A few days after new moon, in the hour after sunset, the sky's most ordinary wonder is available to anyone who looks up. The crescent Moon hangs low over the sunset glow, a bright sliver of the lunar disk lit by direct sun. And around that sliver, faint but unmistakable once noticed, the rest of the Moon's round face is visible too. It is not black. It is a dim, gray-blue ghost of a full moon, glowing softly against the twilight. Old almanacs called it the old Moon in the new Moon's arms. Modern astronomy names the light that fills it earthshine, and the same light falling anywhere in the universe, on a moon, on a spacecraft, on a night sky, is called earthlight.

The explanation is simple to state and rich in consequence. The Earth, seen from the Moon, is a brilliant object, a blue and white half globe that reflects sunlight the way the Moon reflects it to us, only better. The Earth is both larger and more reflective than the Moon. Its oceans, clouds, ice sheets, and vegetation send back a substantial fraction of the sunlight that strikes them, and some of that reflected light travels the quarter million miles to the Moon. There it falls on the lunar night, on the part of the Moon not facing the Sun, and makes the dark portion of our companion faintly visible from Earth. We see, in the Moon's dusk, our own planet's reflection.

The geometry is what makes the sight so reliable. When the Moon is a thin crescent from Earth, the Earth is nearly full as seen from the Moon. The two bodies run opposite in phase. On the evenings when the lunar crescent is at its thinnest, the Earthlight falling on the lunar night is at its strongest, so the effect is at its best precisely when the direct sunlight on the Moon is at its weakest. The faint full disk between the crescent's horns is therefore not a rare accident but a scheduled event, repeating every month in the days around new moon, waiting for any observer with a clear western horizon and a moment to spare.

Earthshine matters far beyond its beauty. The Moon, in those crescent evenings, is a screen placed in the sky, and the light on the screen is a sample of the whole Earth at once. Every ocean, every cloud field, every forest and desert and ice sheet contributes to it. By measuring the brightness and color and polarization of earthshine, astronomers can compute the reflectivity of the entire planet, the quantity climate scientists call the Earth's albedo. That single number, the fraction of incoming sunlight the Earth sends back to space, is a first-order control on the planet's temperature, and earthshine provides one of the few ways to watch it change over the years.

The same measurements run the tape of discovery backward and forward. Backward, because the Moon's faint glow helped early astronomers reason their way to the true nature of the Earth Moon system, from Leonardo da Vinci's first correct account to the careful brightness studies of the nineteenth century. Forward, because when astronomers find an Earth sized planet orbiting a distant star, the light they will analyze is in essence earthshine from another world. The techniques now refined on our own planet's reflection, the spectral fingerprints of oxygen, ozone, water, and vegetation, are the same techniques that will one day answer whether other Earths exist and whether any of them are alive.

This book is about that light: what it is, how it was discovered and explained,

The subject is friendly to every level of attention. A reader with no equipment and ten minutes can see earthshine, tonight or any clear crescent. A reader with binoculars can study it. A reader with a camera can measure it, roughly but genuinely, against standard stars. A reader with a lifetime can add to its climate record. This range, from sidewalk to science without a break, is rare, and the book keeps returning to it, because it is not a decoration of the subject but part of its structure: the glow is easy enough to have always been known, and information rich enough to still be at the frontier. how it is observed and measured, and what it tells us about the planet that makes it. The chapters move from the sight itself, familiar to every sky watcher, through the physics of double reflection, to the instruments and programs that turned an evening's glow into a scientific record, and finally to the uses of earthshine in climate science and in the search for habitable worlds. There is no light in the night sky easier to see, and few that carry as much information.

## Chapter 1: The Discovery of Earthshine and Earthlight

The glow was seen long before it was understood. Ancient observers noticed the whole face of the Moon at crescent phases, and some treated the dim full disk as its own kind of light, a lunar property rather than a borrowed one. The dim full face of the crescent phase appears in weather lore and almanac tradition across Europe, usually as an omen: the old Moon holding the new, a saying that occasionally meant fair weather coming and just as occasionally meant the drowning of sailors, depending on which almanac a reader trusted. The line of poetry most often attached to the sight comes from a Scots ballad about a doomed sea voyage, in which the narrator sees the

new Moon with the old Moon in her arm, a phrase that has kept company with the phenomenon for centuries precisely because it captures the observation so plainly.

The correct explanation arrived with the notebooks of Leonardo da Vinci

Before Leonardo, the guesses ranged widely, and they make a small museum of early ideas about the sky. Some traditions assigned the glow to the Moon's own faint luminosity, a kind of native phosphorescence, and the idea persisted in corners for centuries after the correct explanation was in print. Others treated the dark face as translucent, lit faintly by sunlight passing through the Moon's body, which at least respected the observation that the glow tracked the Sun. What the museum shows is not foolishness but the cost of a missing premise: until the Earth was allowed to be a reflecting body like any other, the explanation could not close, and the glow kept waiting for the idea of the Earth that the telescope era would finally supply. around the beginning of the sixteenth century. In pages filled with studies of light, optics, and the Moon's appearance, Leonardo argued that the ghostly glow on the Moon's dark side is sunlight reflected from the Earth. The Earth, he wrote, with its seas acting as mirrors, sends light to the Moon just as the Moon sends light to us. He got the geometry right and the mechanism right, and he even reasoned, correctly, that the brightness of the lunar night must depend on the character of the Earth's surface, an idea that would not be tested with instruments for another four hundred years. It is a remarkable piece of thinking for a man who never saw a telescope.

Galileo Galilei, a century later, reached the same conclusion in print and tied it to his telescopic view of the Earth as a body like the Moon. In his account of the Earth's reflection he asked the reader to consider the Moon seen from the Earth and the Earth seen from the Moon as mirrors facing one another, each illuminating the other's night. Galileo's phrasing made the idea part of the public case for the Copernican worldview: if the Earth is a reflecting, phase-bearing body like the Moon, the boundary between the earthly and the celestial dissolves.

The seventeenth century added the patient record keeping. Giovanni Battista Riccioli, the Jesuit astronomer of Bologna, kept careful notes of the Moon's appearance and of the faint light on the night side, distinguishing the genuine reflection of earthlight from the scattered glare of twilight. His records of the ashen illumination, made in the 1640s, are among the first systematic observations, and they included the honest uncertainty of a good observer, since the boundary between earthshine and the sky's own glow is exactly the kind of thing an honest astronomer learns to measure rather than assume.

The modern term earthshine settled into astronomy in the nineteenth century

Between the notebooks and the photometers, the eighteenth century kept its own kind of watch. The telescope age had settled the geometry, and the observers who drew the Moon, mapping seas and craters with growing precision, noted the earthlit disk as a matter of course, a useful extra light in which the night side's features could be checked between lunations. Some used the glow deliberately: when a bright comet or a nova needed a reference point near the crescent Moon, the faint face was there, charted and steady, a photometric landmark before photometry had a name. The glow's service in these small observations never made the histories, but it marks the stretch of time when the phenomenon passed from marvel into background knowledge, the fate of all well-explained things., when photometry matured enough to treat the glow as a quantity. Astronomers measured the brightness of the Moon's night side against stars and against the daylight side, and they discovered the first great surprise of the subject: the Earth, as a reflector, is several times brighter than the Moon. The comparison of earthlight and moonlight became a standard exercise, and the numbers that emerged, that a full Earth on the Moon is dozens of times brighter than a full Moon on the Earth, still stand as one of the cleanest ways to feel the difference between the two worlds.

The discovery story is therefore not a single eureka but a long convergence: folklore saw the glow, Leonardo explained it, Galileo publicized it, Riccioli logged it, and the photometrists measured it. Each step brought the same small observation closer to the center of science, until a light visible on any clear crescent evening became a tool for studying the planet that casts it.